

Gouki Minegishi

Ph.D. Candidate

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RESEARCH INTERESTS

Mechanistic Interpretability

In-Context Learning

Large Language Models

Reasoning

Sparse Autoencoders

Neural Network Circuits

EDUCATION

Ph.D. in Computer Science

The University of Tokyo

Advisor: Prof. Yutaka Matsuo

Apr 2025 – Mar 2028

M.S. in Computer Science

The University of Tokyo

Apr 2023 – Mar 2025

B.S. in Engineering

The University of Tokyo

Apr 2019 – Mar 2023

SELECTED PUBLICATIONS

1

Mechanism of Task-oriented Information Removal in In-context Learning

Hakaze Cho, Haolin Yang, [Gouki Minegishi](#), and Naoya Inoue

[International Conference on Machine Learning](#) [ICLR 2026](#)

[arXiv](#)

2

RL Squeezes, SFT Expands: A Comparative Study of Reasoning LLMs

Kohsei Matsutani, Shota Takashiro, [Gouki Minegishi](#), Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo

[International Conference on Machine Learning](#) [ICLR 2026](#)

[arXiv](#)

3

Topology of Reasoning: Understanding Large Reasoning Models through Reasoning Graph Properties

[Gouki Minegishi](#), Hiroki Furuta, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo

[Neural Information Processing Systems](#) [NeurIPS 2025](#)

[arXiv](#)[Code](#)

4

Beyond Induction Heads: In-Context Meta Learning Induces Multi-Phase Circuit Emergence

[Gouki Minegishi](#), Hiroki Furuta, Shohei Taniguchi, Yusuke Iwasawa, and Yutaka Matsuo

[International Conference on Machine Learning](#) [ICML 2025](#)

[OpenReview](#)[arXiv](#)[Code](#)

5

Rethinking Evaluation of Sparse Autoencoders through the Representation of Polysemous Words

[Gouki Minegishi](#), Hiroki Furuta, Yusuke Iwasawa, and Yutaka Matsuo

[International Conference on Learning Representations](#) [ICLR 2025](#)

[OpenReview](#)[arXiv](#)[Code](#)

6

Interpreting Multi-Attribute Confounding through Numerical Attributes in Large Language Models

Hirohane Takagi*, [Gouki Minegishi*](#), Shota Kizawa, Issey Sueda, Hitomi Yanaka (* *Equal Contribution*)

[International Joint Conference on NLP / Asia-Pacific Chapter of ACL](#) [IJCNLP-AAACL 2025](#)

[ACL Anthology](#)[arXiv](#)

7

Bridging Lottery Ticket and Grokking: Understanding Grokking from Inner Structure of Networks

[Gouki Minegishi](#), Yusuke Iwasawa, and Yutaka Matsuo

[Transactions on Machine Learning Research](#) [TMLR](#)

[arXiv](#)[Code](#)

8

Towards Empirical Interpretation of Internal Circuits and Properties in Grokked Transformers on Modular Polynomials

Hiroki Furuta, [Gouki Minegishi](#), Yusuke Iwasawa, and Yutaka Matsuo

[Transactions on Machine Learning Research](#)

TMLR

arXiv

9

ADOPT: Modified Adam Can Converge with Any β_2 with the Optimal Rate

Shohei Taniguchi, Keno Harada, [Gouki Minegishi](#), et al.

[Neural Information Processing Systems](#)

NeurIPS 2024

OpenReview

arXiv

Code

RESEARCH & WORK EXPERIENCE

Research Scientist

Jun 2025 – Present

Third Intelligence, Inc.

Research on developing new forms of intelligence beyond conventional AI, aiming to bridge human-like reasoning, perception, and action.

Chief AI Engineer

Nov 2024 – May 2025

Matsuo Institute, Inc.

Developed automated driving software, including 3D object detection with deep learning and integration of the World Model into Automated Driving.

Research Intern

Aug 2024 – Oct 2024

Preferred Networks, Inc.

Research on in-context learning in foundation models, with knowledge conflicts.

ACADEMIC HONORS

- ★ Young Researcher Encouragement Award, NLP 2025
- ★ Outstanding Presentation Award, JSAI 2024
- ★ Oral Presentation at ICLR 2025 Workshop (Building Trust in LMs)
- ★ Oral Presentation at ICLR 2024 Workshop (Bridging Practice and Theory)
- ★ Selected for "BOOST NAIS" Scholarship Program for fostering advanced AI talents leading the next-generation intelligent society

ACADEMIC SERVICE

- Reviewer: ICLR 2026, AAAI 2026, NeurIPS 2025, ICML 2025/2026
- Organizer: Mechanistic Interpretability Organized Session at JSAI 2025